INCH-POUND

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DETAIL SPECIFICATION

CONNECTORS, PLUG, AND RECEPTACLE (ELECTRICAL, AUDIO, WEATHERPROOF, TEN CONTACT, POLARIZED)

Inactive for new design after 17 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

Scope. This specification covers waterproof, polarized, ten contact, electrical connectors (see 6.3) (plugs and receptacles) designated (see 6.4) as follows:

Designation	Description
U-77( )/U	Connector, Plug, (Rigid Contacts, mates with all non-rigid contact plugs and receptacles)
U-78( )/U	Connector, Plug, (Non-rigid contacts, mates with all rigid contact plugs and receptacles)
U-79( )/U	Receptacle Connector, (Non-rigid contacts, mates with all rigid contact plugs)
U-126( )/U	Connector, Receptacle, (Rigid contacts, mates with all non-rigid contact plugs)
U-127( )/U	Connector, Plug, (Rigid contacts, right angle, mates with U-79( )/U and U-165( )/U connectors)
U-128( )/U	Connector, Plug, (Non-rigid contacts, right angle, mates with U-126( )/U connector and all receptacles)
U-161( )/U	Connector, Plug, Electrical, (Rigid contacts, mates with all non-rigid contact plugs and receptacles)
U-162( )/U	Connector, Plug, Electrical, (Non-rigid contacts, mates with all rigid contact plugs and receptacles)
U-163( )/U	Connector, Plug, Electrical, (Rigid contacts, right angle, mates with U-79( )/U and U-165( )/U connectors)
U-164( )/U	Connector, Plug, Electrical, (Non-rigid contacts, right angle, mates with U-126( )/U- connector)
U-165( )/U	Connector, Receptacle, Electrical, (Non-rigid contacts, mates with all rigid contact plugs)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus OH 43218-3990 or email to RFConnectors@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications and standards, and handbooks. The following specifications and standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-14072 Finishes for Ground Based Electronic Equipment
MIL-DTL-45204 Gold Plating, Electrodeposited

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202-101 Salt Atmosphere (Corrosion)
MIL-STD-202-107 Thermal Shock
MIL-STD-202-201 Vibration
MIL-STD-810 Environmental Engineering Considerations and Laboratory Tests
MIL-STD-13231 Marking of Electronic Items

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ECOM DRAWINGS (SIGNAL CORPS)

ECOM DWG SC-A-46439	List of Accessories for Package Tester
ECOM DWG SC-DL-22728	Connector, Receptacle, Electrical, U-165( )/U
ECOM DWG SC-DL-34036	Connector, Plug, Electrical, U-161( )/U
ECOM DWG SC-DL-34037	Connector, Plug, Electrical, U-162( )/U
ECOM DWG SC-DL-34038	Connector, Plug, Electrical, U-163( )/U
ECOM DWG SC-DL-34039	Connector, Plug, Electrical, U-164( )/U
ECOM DWG SC-DL-68417	Connector, Plug, U-77( )/U
ECOM DWG SC-DL-83049	Receptacle Connector, U-79( )/U
ECOM DWG SC-DL-99587	Connector, Plug, U-78( )/U
ECOM DWG SC-DL-99590	Connector, Receptacle, U-126( )/U
ECOM DWG SC-DL-105995	Connector, Plug, U-127( )/U
ECOM DWG SC-DL-106000	Connector, Plug, U-128( )/U

(Copies of these documents are available online at DSCC.cddwgs@dla.mil).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM-B488 Gold For Engineering Uses, Electrodeposited Coatings Of

(Copies of these documents are available from http://www.astm.org).

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AMS-QQ-N-290 Nickel Plating (Electrodeposited)

(Copies of these documents are available online from http://www.sae.org).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction. The connectors shall be constructed in accordance with the following except the contact and terminal finish shall be in accordance with 3.5.

Connector, Plug,	U-77( )/U ECOM DWG SC-DL-68417
Connector, Plug,	U-78( )/U ECOM DWG SC-DL-99587
Connector, Plug,	U-127( )/U ECOM DWG SC-DL-105995
Connector, Plug,	U-128( )/U ECOM DWG SC-DL-106000
Connector, Plug,	U-161( )/U ECOM DWG SC-DL-34036
Connector, Plug,	U-162( )/U ECOM DWG SC-DL-34037
Connector, Plug,	U-163( )/U ECOM DWG SC-DL-34038
Connector, Plug,	U-164( )/U ECOM DWG SC-DL-34039
Connector, Receptacle,	U-79( )/U ECOM DWG SC-DL-83049
Connector, Receptacle,	U-126( )/U ECOM DWG SC-DL-99590
Connector, Receptacle,	U-164( )/U ECOM DWG SC-DL-22728

3.2 Preproduction samples. The contractor shall furnish the following preproduction samples for each type of connector on order for approval provided the invitation for bids and contract require samples (see 4.4):

18 assembled connectors
1 unassembled connector
6 applicable mating connectors

3.3 Castings. Castings shall be of uniform quality and condition and free from cracks, harmful shrinkage, porosity, gas holes, foreign matter, and other injurious defects. The surface of the castings shall be free from pits, parting lines, porous areas, fins, ridges, modules, raised metal, and scale. All castings shall be completely cleaned prior to presentation for inspection. Castings shall not be plugged or welded, nor shall imperfections be filled in.

3.4 Cleaning.

3.4.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely.

3.4.2 Connectors. After assembly, connectors shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the connectors.

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3.5 Finish. These connectors shall be finished in accordance with its drawings (see 3.1) and as specified for each in accordance MIL-DTL-14072 as applicable to each connector, (see 4.5), except that the electrical contacts and wiring terminals shall be gold plated to a minimum thickness of 30 microinches, in accordance with MIL-DTL-45204, type II, grade C, class0 or ASTM-B488, type II, code C class .75, over a 50 microinches minimum nickel underplate, in accordance with SAE-AMS-QQ-N-290, class 1, over copper flash. This requirement supersedes the contact and terminal plating requirements of the related SC drawings. Terminals maybe pretinned prior to shipment.

3.6 Marking. Marking shall conform to Specification MIL-STD-13231. (See 4.5)

3.7 Interchangeability. Like connectors and subassemblies of like connectors shall be physically and functionally interchangeable without modification of such items. (See 4.7) Individual parts shall not be hand-picked for fit.

3.8 Contact depression. The force required to depress the contacts of connectors with non-rigid contacts to a distance of 0.135 inch from the front edge of the shell shall not exceed 25 pounds. The contact resistance (see 3.9) shall not change. (See 4.8)

3.9 Contact resistance. The electrical resistance of each non-rigid contact shall be such that the terminal-to-terminal resistance of mated connector contacts shall not exceed 0.015 ohm (see 4.9).

3.10 Dielectric strength. The connectors shall show no evidence of breakdown when subjected to a potential of 500 volts rms, 60 cycles per second, for a minimum of one minute. (See 4.10 and 4.10.1)

3.11 Insulation resistance. The insulation resistance shall be not less than 1000 megohms except for unmated connectors following the immersion test when it shall be not less than 100 megohms. (See 4.11)

3.12 Air pressure. The connector shall show no leakage when subjected to pressure of 2.5 psi applied to the contact face and then to the rear of the connector. (See 4.12)

3.13 Bounce. After being tested as specified in 4.13, there shall be no evidence of loose parts or physical damage other than surface abrasion.

3.14 Drop. After being tested as specified in 4.14, the connectors shall show no evidence of physical damage which affects matability nor shall there be any loose parts.

3.15 Durability. Connectors shall show no mechanical damage and shall meet the requirements specified for contact resistance, dielectric strength, and air pressure after being tested as specified in 4.15.

3.16 Immersion. Following immersion as specified in 4.16, there shall be no evidence of leakage into the body of unmated connectors or on the contact face area of mated connectors and the connectors shall meet the requirement specified for insulation resistance.

3.17 Temperature, relative humidity. After being tested as specified in 4.17, connectors shall have an insulation resistance of not less than 100 megohms.

3.18 Pull. Mated connectors shall with stand an axial pull of less than 40 pounds applied to the shell and 25 pounds applied to the cable. The force shall be applied abruptly. (See 4.18)

3.19 Corrosion. There shall be no evidence of base metal corrosion when the connectors are tested as specified in 4.19.

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3.20 Thermal shock. At the extreme temperatures during the test specified in 4.20, the connectors shall be capable of being mated and unmated. The contact resistance of connectors with non-rigid contacts shall be 5.0 ohms or less at the low extreme. After the test, the contact resistance of connectors with non-rigid contacts shall be as specified in 3.9.

3.21 Vibration. When mated connectors are tested as specified in 4.21, there shall be no evidence of cracking, breaking, or loosening of parts, and the plug shall not become disengaged from the receptacle. The connectors shall meet the requirements specified for contact resistance, dielectric strength and insulation resistance following the test.

3.22 Workmanship. Connectors and associated fittings shall be processed in such a manner as to be uniform in quality and shall be free from sharp edges, burrs and other defects that will affect life, serviceability or appearance.

3.23 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Contractors responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the exanimation and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).

b. Conformance inspection (See 4.4 and 4.5).

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4.3 Preproduction Inspection. This inspection may be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (see tables II, III, and IV, respectively). The preproduction inspection will normally be performed in this order (1) vibration, (2) bounce, (3) shock, drop, and (4) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing (see 3.2).

Table I. Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Requirement	Test method
Bounce	3.13	4.12
Drop	3.14	4.13
Temperature, relative humidity	3.17	4.16
Thermal shock	3.20	4.19

4.4 Conformance inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the conformance inspection:

Item	Where required
Finish	3.5
Marking	3.6

4.5 Conformance inspection of equipment before preparation for delivery. The contractor, to demonstrate compliance with specific requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government at its discretion may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6 6) Test equipment for Government verification inspection shall be made available by the contractor.

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4.5.1 Group A inspection. Group A inspection shall consist of the inspection specified in table II in the order shown.

4.5.1.1 Sampling plan (group A). Table II tests shall be performed on a production lot basis. Samples shall be selected in accordance with table III. If one or more defects are found, the lot shall be screened for that particular defect and defective parts removed. A new sample of parts shall be selected in accordance with table III and all group A tests again performed. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification. When defects have no effect on the use or performance of the subject parts, reworking of the parts may be permitted upon approval of the Government. Such allowances are infrequent and only apply to particular circumstances.

Table II. Group A inspection

Inspection	Requirement	Teat method
Visual and Mechanical	3.22	4.22
Construction	3.1	
Castings	3.3	
Cleaning	3.4	
Finish	3.5	
Marking	3.6	
Interchangeability	3.7	4.6
Electrical		
Dielectric Strength	3.10	4.9
Insulation Resistance	3.11	4.10
Air-pressure	3.12	4.11

*All electrical and air-pressure defects are considered major.

TABLE III. Inspection level.

Lot size	Visual and mechanical inspection
1 to 19	All
20 to 280	20
281 to 1,200	47
1,201 to 3,200	53
3,201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143

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4.5.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV in the order shown, and shall be made on sample units which have been subjected to and passed the group A inspection. Connectors having identical piece parts may be combined for lot purposes and shall be in proportion to the quantity of each PIN numbered connector produced.

4.5.2.1 Group B sampling plan. A sample of parts shall be randomly selected in accordance with table V. If one or more defects are found, the lot shall be screened for that particular defect and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected and subjected to all tests in accordance with table IV. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

4.5.2.2 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted. Any connector deformed or otherwise damaged during testing shall not be delivered on the contract or order.

Table IV. Group B inspection

Inspection	Requirement	Test method
Contact depression	3.8	4.7
Contact resistance	3.9	4.8
High altitude breakdown	3.10	4.9.1
Pull	3.18	4.17

TABLE V. Inspection level.

Lot size	Visual and mechanical inspection
1 to 19	All
20 to 280	20
281 to 1,200	47
1,201 to 3,200	53
3,201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143

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4.5.3 Group C inspection. Group C inspection shall consist of the inspections specified in table VI, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection..

4.5.3.1 Sampling plan. Twelve sample units of the same PIN shall be selected from the lot.

4.5.3.2 Failures. If one or more sample units fails to pass group C inspection, the lot shall be considered to have failed.

4.5.3.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or order.

Table VI. Group C inspection

Inspection	Requirement	Test method
Durability	3.15	4.14*
Immersion	3.16	4.15
Corrosion	3.19	4.18*
Vibration	3.21	4.20

*Sample units subjected to this inspection shall not be furnished on contract.

4.5.3.4 Noncompliance. If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5)

4.6 Interchangeability. The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) Angular locatian of bayonet pins.
(b) Diameter of bayonet pins.
(c) Outside diameter of shell.
(d) Location of key-way with respect to bayonet pin.
(e) Distance between face of rigid contacts and bottom of bayonet pins.
(f) Inside diameter of bayonet pin circle.
(g) Diameter of rubber seal.
(h) Dimension between top of rubber seal and bottom of bayonet pins.
(i) location of contact “A” In reference to bayonet pin.
(j) location of J-slot.
(k) Dimension between the front face and final position depth of shell.
(l) Dimension between high point in J-slot and face of shell.
(m) Width of J-slots.
(n) Inside diameter of shell.
(o) location of contact ‘A” with final position of J-slot.

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4.7 Contact depression. A plate shall be applied to the contacts of connectors with non-rigid contacts in such a manner that the normal plane of the contacts is depressed to a distance of 0.135 inch from the front edge of the shell. The depressive force shall be measured and shall conform the requirement specified in 3.8. The inspection specified for contact resistance shall be performed after completion of the above procedure.

4.8 Contact resistance. The contact resistance of a connector with non-rigid contacts mated with a rigid contact connector or equivalent test jig shall be measured with a Kelvin bridge or other approved method. (See 3.9)

4.9 Dielectric strength. A potential of 500 volts rms shall be applied between each contact of the connector and the remaining contacts connected together and to the shell. The voltage shall be increased gradually from zero to 500 volts within 5 seconds and shall be maintained at 500 volts for approximately but not less than one minute. (See 3.10)

4.9.1 High altitude breakdown. The connector shall be placed in an altitude chamber and the barometric pressure reduced to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). After temperature and pressure stabilization, the test specified in 4.9 shall be repeated.

4.10 Insulation resistance. The insulation resistance shall be measured between each contact of the connector and the remaining contacts connected together and to the shell. (See 3.11)

4.11 Air pressure. A pressure of 2.5 pounds per square inch (psi) shall be applied to the contact face and then to the rear of the connector, using the differential leakage tester described in Figure 1, or an equivalent method approved by the Government, to detect leakage.

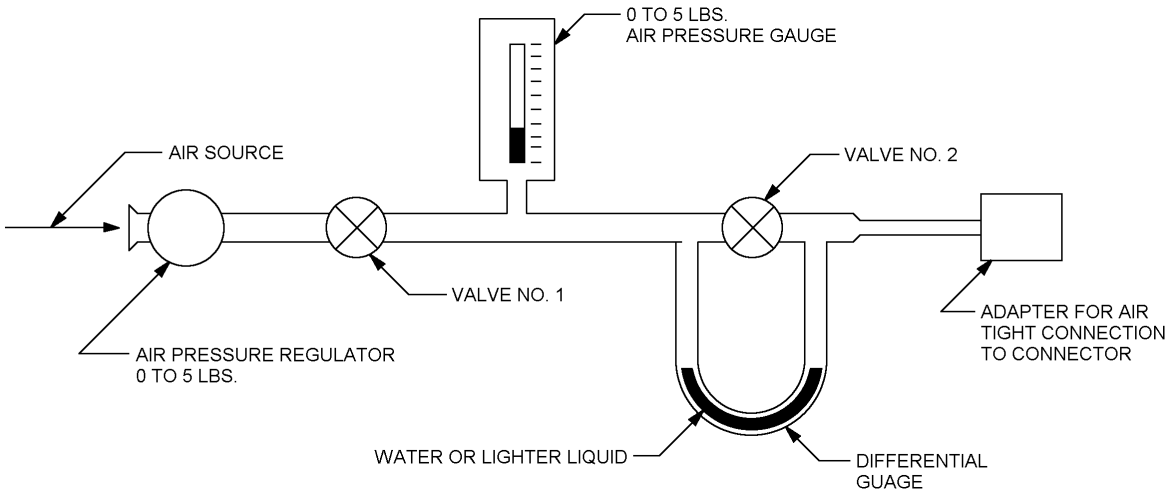


Figure 1. Diagram of air pressure test set-up.

4.11.1 Test method. The connector shall be attached to the leakage tester with its face against the appropriate jig of the leakage tester. With valve no 2 open, valve No. 1 and regulator shall be adjusted to supply 2.5 psi pressure to the connector face. After at least 15 seconds, valve No. 1 shall be closed. Then valve No. 2 shall be closed and the gage observed at least 30 seconds for a continuous change in the level of the two columns (a small monetary change when valve No. 2 is closed is not evidence of leakage). Valve No. 2 shall then be opened before removal of the connector. The above procedure shall be repeated with the rear of the connector attached to the proper jig of the tester.

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4.12 Bounce test (see 3.13). The connectors shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

a. Cover the tester bed with a panel of .500 inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

b. Place the connectors on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3-inches and not less than 1-inch.

c. Operate the package tester, shafts in phase, for a total of 3 hours at 284 ± 2 rpm.

d. Following the above procedure, the connectors shall be tested for contact resistance, (4.9), insulation resistance (4.11), dielectric strength (4.10) and air pressure (4.12).

4.13 Drop (see 3.14). The connectors shall be dropped at random six times from a height of six feet onto a 2 inch fir floor or barrier backed by a concrete or rigid steel frame. The connectors shall then be subjected to the tests specified for contact resistance (4.9), dielectric strength (4.10), insulation resistance (4.11) and air pressure (4.12).

4.14 Durability. The connector shall be subjected to 1500 cycles of mating and unmating (a cycle consisting of mating, locking, unlocking, and complete separation of the connectors) at the rate of not more than 30 cycles per minute (see 3.15). The connector shall be examined for mechanical damage and then subjected to the tests specified for contact resistance (4.9) dielectric strength (4.10) and air pressure (4.12). Failure of any of these tests shall constitute failure of the durability test.

4.15 Immersion. To determine compliance with the requirement specified in 3.16, the connectors shall be immersed in tap water to a depth of six (6) feet for a period of 48 hours in accordance with the following:
Connector plug shall be assembled to test cables (Cable, Special Purpose, WM-70B/U). The length of cable shall be such that the cables extend a few feet outside of the tank.
Connector receptacle shall be sealed completely against leakage or shall be mounted by their normal means to the wall of the tank so that the terminal end of the shell is outside of the tank.
Fifty percent of the connectors tested shall be mated. Insulation resistance (see 4.10) of the still mated connectors shall be measured at the end of the test and shall meet the requirement specified in 3.11.
Fifty percent of the connectors tested shall be unmated. Upon completion of the test, all excess moisture shall be removed and the connector dried by room temperature compressed air for a period of 5 minutes. Insulation resistance (see 4.11) shall be measured within 30 minutes after the connector is removed from the water and shall meet the requirement specified in 3.11.
All connectors shall be inspected for internal leakage after being removed from the water.

4.16 Temperature, relative humidity. Connectors shall be subjected to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time shall conform to 520.2 of Standard MIL-STD-810. The connectors shall then be removed from the humidity chamber and allowed to dry for a period of approximately 24 hours at 25º ± 5ºC, with the relative humidity controlled at 50 ± 5 percent. The connectors shall be the subjected to the test specified for insulation resistance in 4.11. (See 3.17)

4.17 Pull (see 3.18). The receptacle shall be mounted rigidly in the vertical axis with the mating face downward. The plug, assembled to cable specified in 4.15(a) with a device such as a sling or harness affixed to the plug, shall be mated with the receptacle. A dead weight of 40 pounds shall be placed abruptly on the device. Another device shall be attached to the cable and 25 pounds shall be placed abruptly on it.

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4.18 Corrosion (see 3.19). Connectors shall be tested in accordance with MIL-STD-202-101. The following details and exceptions shall apply:

a. Test condition: B
b. Salt solution: 5 percent.

4.19 Thermal shock (see 3.20). Connectors shall be subjected to temperature cycling in accordance with MIL-STD-202-107 Test Condition D. Half of the connectors tested shall be mated. At the extreme temperatures, the connector shall be mated and unmated and the contact resistance measured to determine compliance with the requirement specified in 3.20.

4.20 Vibration. Connectors, mated as follows, shall be subjected to the vibration in accordance with MIL-STD-202-201.

a. Receptacles shall be mounted securely to a plate mounted on the vibration table. The corresponding mating plugs shall then be mated to the receptacles.

b. Plugs with non-rigid contacts shall be securely mounted by metal straps to a plate mounted on the vibration table in such a manner that the mating plug, when mated with it, is not supported by any means other than the coupling. The corresponding plug with rigid contact shall be mated with the non-rigid plug.

Following the vibration procedure, the connectors shall be subjected to the tests specified for contact resistance (4.9), dielectric strength (4.10) and Insulation resistance (4.11) to determine compliance with requirement specified in 3.21.

4.21 Visual and mechanical examination. Connectors shall be examined to verify that the design, construction, physical dimensions, assembly instructions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.5, 3.6 and 3.22.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packing requirements. Packing requirements are maintained by the Inventory Control Point's packing activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Connectors covered by this specification are intended for use in audio frequency circuits at 60 millivolts minimum to 60 volts maximum potential and 0.5 amperes maximum current. Applications are such as bandset, headset, and microphone operation. Circuit operation of the contacts can be as follows:

A – Phone
B – Phone return (Ground)
C – Microphone
D – Microphone return (wire application)
E – Microphone return through switch (Ground)
F – Control
H – Control return (Ground)
J – Remote power control
K – Relay to ground
L – Loudspeaker

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.
b. Type required.
c. Preproduction pack(s) as follows:
   Makeup of pack(s)
   Number of each kind of pack to be submitted.
   Inspection to be performed thereon.
d. Marking and shipping of samples
e. Place of final inspection

6.3 Definitions.

6.3.1 Connector. A generic term used to denote an electrical plug or receptacle.

6.3.2 Plug. A connector normally attached to a free-swinging electrical cable.

6.3.3 Receptacle. A connector which is normally rigidly attached to or is an integral part of a supporting surface.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example U-77W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.5 Subject term (keyword) listing.
Castings
Interchangeable
Electrical resistance
Corrosion
Temperatures
Immersion

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians
Army - CR
DLA - CC

Preparing activity
DLA - CC

(Project 5935-2024-004)

Review activities:
Army - AV, EA, MI
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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